

Juniorprogrammierer.de

Python 6: Functions

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What is a function

- A function is a block of code which only runs when it is called.
- A function can return data as a result.
- A function helps avoiding code repetition.
- They can be called multiple times
- Order is important
- Functions are verbs!

Function.py •

Explanation > Function.py > ...

```
1  #calling my_function() here will not work
2
3  def my_function():
4      print("Hello from a function")
5
6  def my_function2():
7      print("Hello from a function2")
8
9  my_function()
10 my_function2()
```

Return-function

- It sends data back to the code that called the function
- When it reaches return it stops the execution and returns the result

#Return

```
def addNumbers(number1, number2):  
    return number1 + number2  
  
result = addNumbers(1,2)
```

Placeholder-function

- Functions cannot be empty this is why there is a placeholder “pass”
- Do you know examples why you need this?

```
#Placeholder  
def myPlaceholderFunction():  
    pass  
  
print(myPlaceholderFunction)
```

Public and private

- Python does not have built-in access modifiers like public or private, it relies on naming conventions and name mangling to indicate visibility and access intent
- By default everything is public
- By starting a variable or method with “_” it is marked as private
- By now you see java seems to have a harder annotation, what a the advantages to learn this anyway first?

```
#Public and private
def public_method(self):
    print("This is a public method.")

def _private_method(self):
    print("This is a private method.")
```

Importing parameter

- A Function can have 1-n importingparameter
- They can be positional or key/value arguments
- You can enforce positional with “/”
- You can enforce key/value with “*”
- You can define default importingparameter => Do you know a usecase why it is a good thing?

```
61  #1 to n importingparameter are possible
62  #you can pass it directly or via "="
63  def my_function(animal, name, age):
64      | print("I have a", age, "year old", animal, "named", name)
65
66  my_function("dog", name = "Buddy", age = 5)
67
68  #Default importingparameter
69  def myFriend(name = "friend"):
70      | print("Hello", name)
71
72  myFriend("Emil")
73  myFriend()
74
75  #Positional only argument
76  def my_function(name, /):
77      | print("Hello", name)
78
79  my_function("Emil")
80  #my_Function(name="Emil") => not allowed
81
82  #Other way around are only Keyword arguments
83  def my_function(*, name):
84      | print("Hello", name)
85
86  my_function(name = "Emil")
```

Do you need a main-methode

- In Python, a main() method is not required for running functions or starting a program. Python scripts are executed from top to bottom
- By using a main() function and the special if __name__ == "__main__": check, you can ensure that certain parts of the code are executed only when the script is run directly, and not when it is imported as a module.

```
#no main method
print("Hello, World!")

def greet():
    print("Welcome to Python!")

greet()

#with main
def greet():
    print("Welcome to Python!")

def main():
    print("Hello, World!")
    greet()

# Ensures that the code runs only when
# the script is executed directly.
if __name__ == "__main__":
    main()
```

Exercise

To translate Fahrenheit to Celsius you need to follow a complicated mathematical formula (see this example)

```
temp1 = 77  
celsius1 = (temp1 - 32) * 5 / 9  
print(celsius1)
```

- Create a fitting function out of this!